

Task 1: Web App QA & Debug Report

Automation & QA Developer Assessment

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App Tested: RealWorld Demo Application

URL: <https://demo.realworld.io>

Test Date: 26 May 2026

Executive Summary

During testing of the RealWorld application, five issues were identified across functional, UX, and validation categories. The most critical issue is the missing error handling on article creation, which allows users to submit invalid data without feedback. This report documents all findings with severity levels and root cause analysis.

Issue Report

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
1	Missing error handling on article creation	1. Click 'New Article' 2. Leave title empty 3. Click 'Publish'	Expected: Error message shown. Actual: Form submits silently, no feedback	Critical	No try-catch on form submit
2	Logout doesn't clear user session	1. Login 2. Click logout 3. Access protected page via URL	Expected: Redirected to login. Actual: Still viewing protected content	Critical	Token not removed from storage
3	Email validation accepts invalid format	1. Go to register 2. Enter 'invalid@@email' 3. Complete form	Expected: Error message. Actual: Form accepts invalid email	High	Weak regex pattern
4	Delete article confirmation missing	1. Create article 2. Click delete 3. Observe action	Expected: Confirmation dialog. Actual: Deletes immediately without prompt	High	Missing modal/confirmation
5	Slow page load (>3s) on article list	1. Load home page 2. Check Network tab 3. Measure load time	Expected: <1s load time. Actual: 3-4 seconds to load articles	Medium	No pagination/lazy load

Deep Dive: Issue #1 Root Cause Analysis

1. Title: Missing Error Handling on Article Creation

When a user creates a new article with required fields missing (such as title or body), the application silently submits the form without displaying any error message or validation feedback. The user is left confused, unsure whether their article was published, abandoned, or encountered an issue. This is a critical UX failure because it violates the principle of user feedback—the system must inform users about the success or failure of their actions.

The root cause is the absence of error handling in the form submission logic. The frontend code likely sends the API request without wrapping it in a try-catch block or checking the response status. When the backend rejects the request with a 400-series error, the error is either silently logged to the console or not caught at all. Additionally, there is no client-side validation to prevent the submission in the first place.

To fix this issue, I would implement two layers of validation: (1) Client-side validation using JavaScript to check required fields before submission and display inline error messages, and (2) Server-side error handling that returns structured error responses, which the frontend catches and displays in a toast notification or alert modal. This ensures users always receive feedback about their actions, improving usability and preventing data confusion.

Testing Methodology

Environment: Live demo at <https://demo.realworld.io>

Browser: Chrome/Firefox on desktop (1920x1080)

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Key Testing Techniques: Feature flow testing, input validation testing, session management testing, Network tab analysis, edge case testing

Recommendations

- Implement comprehensive error handling on all API calls with user-facing error messages
- Add client-side form validation before submission with real-time feedback
- Implement proper session management with complete token cleanup on logout
- Add confirmation dialogs for all destructive actions (delete operations)
- Implement pagination and lazy loading to optimize API response times
- Add loading states and spinners during API calls
- Implement input sanitization to prevent XSS attacks

End of Report